Executive Hotel Booking EDA Report

Day 15 — Hotel Booking Business Performance Dataset

Dataset: 25,180 rows × 35 columns. After removing 180 duplicate Booking IDs, 25,000 unique bookings were analyzed.

Executive KPI Snapshot

KPI	Value
Unique bookings	25,000
Cancellation rate	72.2%
Estimated revenue	13,384,199.45
Average ADR	140.59
Repeat guest rate	17.3%
Average satisfaction	3.85

Data Quality and Preprocessing

Raw missing values were concentrated in Company_ID (20,729), Agent_ID (304), Meal_Type (245), Satisfaction_Score (164), Children (121), Hotel_Location (106), and ADR (82). There were 180 duplicate rows and 180 duplicate Booking IDs. Text categories were standardized, dates converted to datetime, missing operational fields were explicitly encoded, and duplicate booking IDs were removed for analysis.

Outlier Assessment

IQR screening identified potentially unusual values without automatically deleting them. Lead_Time_Days: 2 flagged; ADR: 360 flagged; Estimated_Revenue: 652 flagged; Total_Nights: 136 flagged. Extreme hotel bookings can be legitimate, so the report recommends monitoring rather than indiscriminate removal.

Key Business Insights

1. Cancellation is a major risk: approximately 72.2% of analyzed bookings are canceled, and longer lead times are associated with cancellation.

2. Resort Hotel is the highest-revenue hotel type, while Goa is the highest-revenue location.

3. Online Ta is the largest revenue-contributing market segment, reinforcing the need for segment-specific commercial strategies.

4. ADR and estimated revenue are positively correlated ($r \approx 0.60$), indicating that pricing and stay economics are central to revenue generation.

5. Customer satisfaction varies with guest behavior, including the number of special requests, making experience metrics relevant to management decisions.

Management Recommendations

1. Strengthen cancellation controls with deposits, proactive confirmations, and risk-based cancellation policies.

2. Develop segment-specific pricing and promotion strategies based on ADR, lead time, and cancellation behavior.

3. Prioritize high-performing hotel types and locations while diagnosing underperforming markets.

4. Use revenue-management techniques that jointly optimize ADR, stay length, seasonality, and segment demand.

5. Evaluate distribution channels using both revenue and cancellation economics, not booking volume alone.

6. Monitor customer satisfaction and special-request fulfillment as operational KPIs alongside financial KPIs.

7. Create a monthly executive dashboard covering bookings, cancellations, ADR, revenue, lead time, segment mix, and satisfaction.

Conclusion

The hotel portfolio's performance is driven by a combination of cancellation risk, market mix, location, pricing, stay behavior, and customer experience. Management should focus first on protecting revenue from avoidable cancellations, then optimize pricing and channel/segment mix while maintaining service quality.